

PHITSANULOK, Thailand

WMO: 483780

Lat: 16.7961N

Lon: 100.2758E

Elev: 46

StdP: 100.78

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.5	(c) 18.0	(d) 10.6	(e) 8.0	(f) 22.6	(g) 12.4	(h) 9.0	(i) 23.5	(j) 5.0	(k) 27.6	(l) 4.4	(m) 28.1	(n) 0.5	(o) 10	(p) 0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.5	38.0	25.6	37.0	25.6	36.0	25.5	27.6	33.1	27.2	32.6	27.0	32.4	2.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.9	29.5	26.1	21.6	29.3	25.7	21.1	29.1	88.2	33.4	86.7	32.7	85.5	32.5	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.0	4.3	3.8	DB	13.8	39.5	2.0	1.2	12.3	40.4	11.1	41.1	10.0	41.8	8.5	42.7	
			WB	12.1	28.9	1.9	0.7	10.7	29.4	9.6	29.8	8.6	30.2	7.2	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	25.4	27.4	29.6	31.3	30.5	29.7	29.0	28.7	28.7	28.5	27.4	25.4
	DBStd	2.40	2.02	1.95	1.91	1.74	1.82	1.23	1.19	1.05	0.97	1.17	1.64	2.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6738	478	487	609	641	636	591	588	578	561	574	521	476
	CDD18.3	3698	220	253	351	391	377	341	330	320	311	316	271	218
	CDH23.3	42508	1967	2666	4321	5380	4928	4190	3793	3582	3454	3503	2777	1948
	CDH26.7	19545	786	1257	2264	3112	2604	1953	1580	1395	1326	1387	1137	744
Wind	WSAvg	1.3	1.1	1.2	1.5	1.6	1.6	1.6	1.4	1.4	1.2	1.1	1.1	1.1
Precipitation	PrecAvg	1349	7	13	31	54	183	174	190	244	257	147	35	9
	PrecMax	1854	72	90	164	195	540	353	466	441	478	370	187	106
	PrecMin	929	0	0	0	0	10	0	66	100	66	6	0	0
	PrecStd	242	15	17	35	45	99	82	75	90	98	86	43	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	35.9	37.9	39.8	39.2	37.0	35.6	34.3	34.2	34.4	34.8	34.0
		MCWB	23.3	23.6	24.2	25.0	26.0	25.7	26.2	26.5	26.5	26.1	25.3	24.3
	2%	DB	32.8	34.8	36.8	38.3	37.7	35.3	34.2	33.4	33.2	33.6	33.7	32.9
		MCWB	23.1	23.9	24.5	25.7	26.2	26.2	26.3	26.3	26.4	25.9	25.0	23.8
	5%	DB	31.9	33.9	35.8	37.3	36.2	34.2	33.2	32.8	32.7	33.0	32.9	31.9
		MCWB	22.7	23.9	24.6	25.7	26.2	26.3	26.2	26.2	26.3	25.7	24.6	23.3
	10%	DB	30.9	32.9	34.9	36.3	35.0	33.2	32.4	31.9	31.9	32.1	31.9	30.8
		MCWB	22.2	23.7	24.7	25.8	26.3	26.3	26.1	26.1	26.2	25.6	24.2	22.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	26.5	27.0	27.7	28.1	27.7	27.5	27.7	27.7	27.5	26.9	26.0
		MCDB	31.1	33.0	34.4	35.5	34.3	32.8	32.5	31.9	31.7	31.9	32.1	30.9
	2%	WB	24.7	25.7	26.4	27.2	27.4	27.2	27.1	27.1	27.2	27.0	26.2	25.1
		MCDB	29.8	32.1	33.8	35.0	33.7	32.3	31.9	31.3	31.2	31.4	30.9	29.9
	5%	WB	24.1	25.2	26.0	26.9	27.0	26.9	26.7	26.7	26.8	26.6	25.8	24.4
		MCDB	29.0	31.4	33.3	34.5	33.3	32.0	31.3	30.8	30.8	30.8	30.2	29.2
	10%	WB	23.5	24.7	25.5	26.5	26.7	26.6	26.4	26.5	26.6	26.3	25.3	23.7
		MCDB	28.4	30.6	32.6	34.0	32.7	31.5	30.9	30.5	30.4	30.2	29.5	28.5
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.4	10.9	10.5	8.9	7.8	7.2	6.8	6.9	7.6	9.2	10.5
		MCDBR	12.2	12.2	11.9	11.4	10.5	9.0	8.2	7.9	7.8	8.6	10.0	10.9
		MCWBR	4.5	4.5	4.1	3.6	3.1	2.7	2.7	2.8	2.8	2.8	3.2	3.7
	5% WB	MCDBR	10.2	10.9	10.9	10.5	9.2	8.0	7.5	7.2	7.3	7.7	8.7	9.7
MCWBR		4.0	4.4	4.0	3.6	3.1	2.9	2.8	3.0	2.8	2.9	3.1	3.5	
Clear-Sky Solar Irradiance	taub		0.488	0.590	0.684	0.627	0.506	0.448	0.437	0.446	0.485	0.552	0.472	0.477
	taud		2.063	1.827	1.649	1.814	2.183	2.367	2.396	2.358	2.234	1.998	2.201	2.115
	Ebn at Noon		795	731	679	722	804	844	854	855	824	754	806	792
	Edh at Noon		159	211	259	220	150	124	120	126	142	175	138	147
All-Sky Solar Radiation	RadAvg		4.69	5.17	5.40	5.96	5.57	5.12	4.80	4.64	4.67	4.75	4.84	4.57
	RadStd		0.27	0.22	0.32	0.28	0.39	0.34	0.28	0.24	0.26	0.28	0.23	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	+0.23	N/A	N/A	N/A	N/A
Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.12	N/A	N/A	+109	+110

Nomenclature: See separate page